#FLG1 GND PWR_FLAG
#FLG2 +24V PWR_FLAG
#FLG3 VSYS PWR_FLAG
#FLG4 DOOR_5V PWR_FLAG

Power

A diode guards against reversed 24 V. The buck feeds the matrix and the logic.

J1 (W08: 24 V in, fuse F2): 1 IN_24V, 2 GND

D1 (1N5822): IN_24V → +24V

C1: +24V — GND, 100uF 35 V

U1 (D36V50F5, 5 V): VIN ← +24V; VOUT → +5V; EN, PG unconnected; GND ← GND

C2: +5V — GND, 470uF 16 V

Pico 2

The diode lets USB and the board's 5 V both feed it.

U2 (Raspberry Pi Pico 2):

UART_TX	GP0	VBUS	+24V
SLIDE_UART	GP1	VSYS	VSYS
GND	GND	GND	GND
SLIDE_STEP	GP2	3V3_EN	
SLIDE_DIR	GP3	3V3	+3V3
SLIDE_EN	GP4	ADC_VREF	
SLIDE_HOME	GP5	GP28	SPARE_GP28
GND	GND	AGND	GND
DOOR_SENSE	GP6	GP27	SPARE_GP27
SPARE_GP7	GP7	GP26	SPARE_GP26
I2C_SDA	GP8	RUN	
I2C_SCL	GP9	GP22	SPARE_GP22
GND	GND	GND	
MATRIX_SCK	GP10	GP21	PORT_AT
MATRIX_TX	GP11	GP20	PORT_SET_3V3
LED1_EN	GP12	GP19	EXPO2
LED2_EN	GP13	GP18	TRIG2
GND	GND	GND	GND
LED3_EN	GP14	GP17	EXPO1
LED4_EN	GP15	GP16	TRIG1

D2 (1N5817): +5V → VSYS

J15 (Spare pins): 1 +3V3, 2 GND, 3 SPARE_GP28, 4 SPARE_GP7, 5 SPARE_GP27, 6 SPARE_GP26, 7 RUN, 8 SPARE_GP22

Filter slider

UART address 0. EN's pull-up keeps it off while the Pico starts.

U9 (BTT TMC2209 V1.3):

SLIDE_EN	EN	VS	+24V
GND	MS1	GND	GND
GND	MS2	A2	MOTOR_A2
SLIDE_UART	PDN_UART	A1	MOTOR_A1
	UART2	B1	MOTOR_B1
	CLK	B2	MOTOR_B2
SLIDE_STEP	STEP	VIO	+3V3
SLIDE_DIR	DIR	GND	GND

C5: +24V — GND, 100uF 35 V

R19: SLIDE_EN — +3V3, 10k

R20: UART_TX — SLIDE_UART, 1k

J10 (W28: slider's motor): 1 MOTOR_A1, 2 MOTOR_A2, 3 MOTOR_B1, 4 MOTOR_B2

J11 (W29: slider's home switch): 1 HOME_SW, 2 GND

R21: HOME_SW — +3V3, 10k

R22: HOME_SW — SLIDE_HOME, 1k

C6: SLIDE_HOME — GND, 100nF

Interlock and LED drivers

The door's switch alone powers U3: door open, every DIM falls to its pull-down.

J2 (W35: door switch): 1 DOOR_FEED, 2 DOOR_5V

F1 (PTC, 0.1 A hold): +5V — DOOR_FEED

C3: DOOR_5V — GND, 100nF

R1: DOOR_5V — DOOR_SENSE, 3.9k

R2: DOOR_SENSE — GND, 6.8k

U3 (74AHCT125):

GND	1OE	VCC	DOOR_5V
LED1_EN	1A	1Y	DIM1
GND	2OE	2Y	DIM2
LED2_EN	2A	3Y	DIM4
GND	3OE	4Y	DIM3
LED4_EN	3A		
GND	4OE		
LED3_EN	4A		
GND	GND		

R3: LED1_EN — GND, 4.7k

R4: DIM1 — GND, 1k

U4 (LDD-1000H): +24V, +24V, GND, GND, DIM ← +VIN, +VIN, −VIN, −VIN; +VOUT, +VOUT, −VOUT, −VOUT → LED1+, LED1+, LED1−, LED1−

J3 (W23: 395 nm): 1 LED1+, 2 LED1−

R5: LED2_EN — GND, 4.7k

R6: DIM2 — GND, 1k

U5 (LDD-1000H): +24V, +24V, GND, GND, DIM2 ← +VIN, +VIN, −VIN, −VIN, DIM; +VOUT, +VOUT, −VOUT, −VOUT → LED2+, LED2+, LED2−, LED2−

J4 (W24: 470 nm): 1 LED2+, 2 LED2−

R7: LED3_EN — GND, 4.7k

R8: DIM3 — GND, 1k

U6 (LDD-1000H): +24V, +24V, GND, GND, DIM3 ← +VIN, +VIN, −VIN, −VIN, DIM; +VOUT, +VOUT, −VOUT, −VOUT → LED3+, LED3+, LED3−, LED3−

J5 (W25: 565 nm): 1 LED3+, 2 LED3−

R9: LED4_EN — GND, 4.7k

R10: DIM4 — GND, 1k

U7 (LDD-1000H): +24V, +24V, GND, GND, DIM4 ← +VIN, +VIN, −VIN, −VIN, DIM; +VOUT, +VOUT, −VOUT, −VOUT → LED4+, LED4+, LED4−, LED4−

J6 (W26: 625 nm): 1 LED4+, 2 LED4−

LED matrix and camera port

5 V signals, through a buffer that's always on.

U8 (74AHCT125):

GND	1OE	VCC	+5V
MATRIX_SCK	1A	1Y	MATRIX_SCK_5V
GND	2OE	2Y	MATRIX_TX_5V
MATRIX_TX	2A	3Y	PORT_SET_5V
GND	3OE	4Y	
PORT_SET_3V3	3A		
+5V	4OE		
GND	4A		
GND	GND		

C4: +5V — GND, 100nF

R11: MATRIX_SCK — GND, 4.7k

R12: MATRIX_SCK_5V — MATRIX_CLK, 47

R13: MATRIX_TX — GND, 4.7k

R14: MATRIX_TX_5V — MATRIX_DATA, 47

R15: PORT_SET_3V3 — GND, 4.7k

R16: PORT_SET_5V — PORT_SET, 47

J7 (W30: LED matrix): 1 +5V, 2 GND, 3 MATRIX_CLK, 4 MATRIX_DATA

J8 (W33: port, DIG I/O 1): 1 PORT_SET, 2 GND

J9 (W34: port, DIG I/O 2): 1 PORT_AT_5V, 2 GND

R17: PORT_AT_5V — PORT_AT, 3.9k

R18: PORT_AT — GND, 6.8k

Cameras

Open-collector triggers, and exposure inputs with pull-ups.

R23: TRIG1 — GND, 4.7k

R24: TRIG1 — TRIG1_BASE, 2.2k

Q1 (2N3904): TRIG1_BASE, GND; TRIG1_OUT

J12 (W31: mono camera): 1 TRIG1_OUT, 2 GND, 3 EXPO1_IN, 4 GND

R25: EXPO1_IN — +3V3, 4.7k

R26: EXPO1_IN — EXPO1, 1k

R27: TRIG2 — GND, 4.7k

R28: TRIG2 — TRIG2_BASE, 2.2k

Q2 (2N3904): TRIG2_BASE, GND; TRIG2_OUT

J13 (W32: colour camera): 1 TRIG2_OUT, 2 GND, 3 EXPO2_IN, 4 GND

R29: EXPO2_IN — +3V3, 4.7k

R30: EXPO2_IN — EXPO2, 1k

LED temperatures

Each thermistor divides 3.3 V against 10k, into the I2C ADC at 0x48.

U10 (ADS1115):

AIN1	AIN0	VDD	+3V3
AIN2	AIN1	SDA	I2C_SDA
AIN3	AIN2	SCL	I2C_SCL
GND	AIN3	ALERT	
GND	ADDR		
	GND		

C7: +3V3 — GND, 100nF

R31: I2C_SDA — +3V3, 4.7k

R32: I2C_SCL — +3V3, 4.7k

J14 (W27: thermistors): 1 TH1, 2 GND, 3 TH2, 4 GND, 5 TH3, 6 GND, 7 TH4, 8 GND

R33: +3V3 — TH1, 10k 1%

R34: TH1 — AIN1, 1k

C8: AIN1 — GND, 100nF

R35: +3V3 — TH2, 10k 1%

R36: TH2 — AIN2, 1k

C9: AIN2 — GND, 100nF

R37: +3V3 — TH3, 10k 1%

R38: TH3 — AIN3, 1k

C10: AIN3 — GND, 100nF

R39: +3V3 — TH4, 10k 1%

R40: TH4 — AIN4, 1k

C11: AIN4 — GND, 100nF

Drawn by electronics/carrier/make.py; change circuit.py, not this file

Sheet: /
File: carrier.kicad_sch

Title: OPTIMA carrier board

Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.8		Id: 1/1